How Should We Chunk Scientific Documents for Retrieval? A Controlled Evaluation of Chunking Strategies for Retrieval-Augmented Generation

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Abstract

Retrieval-augmented generation (RAG) systems split documents into chunks before dense retrieval, yet chunking is chosen largely by convention: fixed-size windows with overlap are the default, and recent work questions whether linguistically or semantically motivated segmentation helps at all. We present a controlled factorial study of five chunking strategies—fixed-token, sentence-window, recursive, structure-aware, and embedding-based semantic—across chunk sizes of 128–512 tokens, overlap levels of 0–20%, two sentence encoders, and BM25 sparse retrieval. Unlike prior evaluations that rely on synthetic or model-judged relevance, we measure document-local retrieval against human-annotated evidence spans in scientific papers (QASPER: 60 papers, 120 practitioner-written questions with expert-annotated evidence spans), with paired Wilcoxon tests, Holm correction, and Cliff's delta effect sizes. Three findings are consistent across metrics and retrievers, and replicate across the two tested sentence encoders. First, chunk size dominates: enlarging chunks from 128 to 512 tokens improves mean nDCG@10 by up to +0.27 for every windowing strategy and both retrievers. Second, 20% overlap never significantly helps any metric and significantly reduces Recall@k in 25 of 108 contrasts. Third, under dense retrieval semantic chunking is the only strategy that beats fixed-size windows—by up to +0.32 nDCG@10 ($\delta = 0.65$) at nominal size 512—while producing an index about 80% smaller. At equal index granularity the picture is more cautious: one matched pair-

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ing is indistinguishable from fixed-token windows and another favors semantic chunking (+0.09, exploratory). Where documents carry strong topical structure, coarse topic-aligned segmentation appears to matter more than boundary placement alone. Findings are stable across encoders: configuration rankings correlate at $\rho = 0.93$ between MiniLM-L6 and BGE-small, and the semantic advantage replicates at every matched size (all $p^{\text{Holm}} \leq 0.016$).

Keywords: Retrieval-augmented generation, dense retrieval, chunking, text segmentation, empirical evaluation, question answering

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1. Introduction

Retrieval-augmented generation (RAG) systems answer questions by first retrieving relevant text and then conditioning a generator on it [1]. In deployed pipelines the retrieved units are not whole documents but *chunks*: contiguous pieces of text produced by splitting each indexed document. How documents are split is one of the first decisions an implementer makes, and one of the least standardized. Fixed-size token windows, overlapping windows, recursive boundary-aware splits, section-preserving splits, and embedding-similarity based "semantic" splits coexist in common tooling, with defaults chosen largely by convention.

The choice matters in principle because chunk boundaries decide where evidence can live. A question about a method's limitation may be answered by two sentences that a naive window places in different chunks; a section-preserving split keeps them together but inflates chunk size, diluting the embedding signal. Prior work confirms that these choices move end-to-end quality [2, 3], yet prior evaluations typically vary one factor at a time on single documents or rely on model-generated relevance judgments [4, 5]. Three gaps remain: to our knowledge, no factorial isolation of size, overlap, strategy, and embedder effects with human chunk-level labels; little use of inferential statistics with corrected significance and effect sizes; and little evidence on long structured scientific documents.

This paper addresses those gaps with a controlled evaluation of five chunking strategies—fixed-token, sentence-window, recursive, structure-aware, and semantic—in a full-factorial design over chunk size (128/256/512 reference tokens) and overlap (0%/20%), instantiated for two dense embedders and BM25 sparse retrieval. We evaluate retrieval quality on QASPER [6], whose questions come with *human-annotated evidence spans* in scientific papers; a

retrieved chunk counts as relevant exactly when it contains annotated evidence. All comparisons use paired per-paper deltas, two-sided Wilcoxon signed-rank tests with Holm correction, and Cliff's delta effect sizes. Specifically, we ask:

RQ1 How does chunk size affect retrieval quality on long scientific documents?

RQ2 Does chunk overlap improve retrieval quality relative to its added indexing cost?

RQ3 Do linguistically informed or embedding-based segmentation strategies outperform fixed-token windowing at matched nominal size?

RQ4 Are observed effects stable across embedding models and across dense versus sparse ranking?

Our contributions are:

1. A pre-specified, fully reproducible factorial study of chunking for retrieval over human-labeled scientific evidence, with all code, filters, and raw results produced by a version-controlled pipeline; the analysis plan was fixed before outcome data were examined, and deviations are documented inline.
2. Evidence that chunk size dominates classical design choices: moving from 128- to 512-token chunks yields large, consistent gains for every windowing strategy, while 20% overlap never helped and often reduced top-k evidence coverage.
3. Evidence, contrary to recent findings on other corpora, that embedding-based semantic segmentation delivers the strongest dense-retrieval quality on long scientific text—with a smaller index, not a larger one—and that this advantage persists across embedders; granularity-matched comparisons, which the analysis plan did not anticipate, temper its interpretation (Section 4.3).
4. Practitioner guidance: tune chunk size first; skip overlap; where documents are topically heterogeneous and retrieval is dense, prefer semantics-aware boundaries, which outperform the windowing heuristics.

Section 2 reviews related work; Section 3 details the methodology; Section 4 presents results by research question; Section 5 discusses mechanisms, implications, and threats to validity; Section 6 concludes.

2. Related Work

2.1. Retrieval-Augmented Generation and Dense Retrieval

Retrieval-augmented generation (RAG) grounds language models in external evidence by retrieving chunks of text and conditioning generation on them [1]. Its retrieval stage descends from dense passage retrieval [7], which embeds queries and candidate texts into a shared vector space, typically via bi-encoders distilled from larger transformers [8]. Because practical corpora are indexed at chunk rather than document granularity, segmenting documents into chunks—"chunking"—precedes embedding and bounds what any retriever can find: evidence split across a poor boundary cannot be retrieved as one coherent unit. Benchmark methodology for such retrieval components is well developed [9], and early question answering over this corpus paired long-context encoders such as Longformer with retrieved evidence [6, 10]; chunking itself, by contrast, has largely been treated as an engineering heuristic rather than an object of controlled study.

2.2. Chunking Strategies in Practice

Practitioner tooling established the default strategy vocabulary. Fixed-token windowing with optional overlap is the baseline; recursive splitting on paragraph, line, sentence, and word boundaries was popularized by LangChain [11]; sentence-window retrieval returns small chunks but expands context around them; embedding-similarity based "semantic" chunking places breakpoints where adjacent-sentence similarity falls below a threshold. Early systematic evidence came from Finardi et al. [2], who varied retrievers, chunk sizes, and generators on a single long narrative document in Brazilian Portuguese, and from Wang et al. [3], who optimized overlapping-window sizes among other RAG hyperparameters. Both observe that chunk configuration materially affects end-to-end quality, supporting our premise, but neither isolates segmentation effects with factorial controls or inferential statistics.

A second wave treats segmentation itself as learnable or model-assisted. Dense X retrieval proposes propositions as the retrieval unit [12]; RAPTOR builds hierarchical tree-organized summaries [13]; LumberChunker prompts an LLM to locate content-shift boundaries in long narratives [14]; Meta-Chunking segments by perplexity-based logical perception [15]; and late chunking embeds the whole document with a long-context encoder and pools token embeddings within boundaries [16]. Industrial guidance such as contextual retrieval prepends document-level context to each chunk before em-

bedding [17]. Surveys chart this widening space [18, 19]. These methods are promising, but their evaluations typically rest on model-judged or synthetic relevance over narrative or web text; none provides factorial, human-labeled evidence on scientific documents - the regime we study.

2.3. Controlled Evaluations of Chunking

Closest to this work are recent controlled evaluations. Qu et al. [20] compare fixed-token chunking against two semantic chunkers across document retrieval, evidence counting, and answer generation, concluding that the gains of semantic chunking rarely justify its cost. Their design, however, evaluates chunkers through proxy tasks on stitched or short-document corpora and measures evidence only as counts within retrieved windows. Wang et al. [4] benchmark segmentation for RAG with LLM-assessed relevance grades; a contemporaneous large-scale study [5] spans 36 methods and reports effectiveness and robustness failures of complex chunkers while likewise relying on model-generated relevance labels. Our study differs in three respects. First, relevance supervision is entirely human: QASPER's evidence spans were annotated by expert readers, eliminating judge-model noise from the measurement instrument. Second, we conduct a full factorial over size, overlap, strategy, and embedder with a pre-registered paired analysis (Holm-corrected Wilcoxon tests plus Cliff's delta effect sizes), whereas prior work compares selected configurations. Third, we target long scientific papers, where section structure is informative and questions demand specific evidence—a regime in which we find, contrary to Qu et al. [20], that semantic chunking delivers the largest gains in the study.

2.4. Evaluation of Retrieval Quality

Standard ranked-retrieval metrics apply unchanged to chunks once chunk-level relevance is defined; we adopt Recall@k, MRR, and nDCG [21]. For statistical practice we follow recommendations to pair observations by evaluation unit, prefer non-parametric tests for bounded metrics, correct for multiplicity, and report ordinal effect sizes alongside significance [22, 23].

3. Methodology

This study asks how chunking strategy choice affects a retriever's ability to locate evidence-bearing chunks in long scientific documents. Section 3.1 formalizes the chunking task; Section 3.2 describes the dataset and

its evidence-based relevance labels; Section 3.3 details the full-factorial design; Section 3.4 specifies the five strategies; Section 3.5, Section 3.6, and Section 3.7 cover retrieval configuration, evaluation metrics, and the preregistered statistical analysis plan. All code, data filters, and raw outputs support exact reproduction and will be archived with a permanent DOI upon acceptance.[1]

Data availability and licensing.. QASPER is distributed under CC BY 4.0 by the Allen Institute for Artificial Intelligence; we redistribute only our filtered manifest (document identifiers, questions, and evidence span offsets), not the paper texts, which are re-downloadable from the official source [6, 24]. Code and result files will be deposited under an open license upon acceptance (see Footnote 1).

3.1. Problem Formulation

Let a document d be a character string assembled from its title and ordered sections. A *chunking configuration* c maps d to an ordered sequence of chunks $\mathcal{C}(d, c) = \{(s_i, e_i)\}_{i=1}^{m}$, where (s_i, e_i) are character offsets into d. Given a query q, a retriever scores every chunk of d and returns a ranking $\pi_q \in \mathcal{S}_m$. Retrieval quality is then judged by whether highly ranked chunks contain the evidence annotated for q. Chunk-level rather than document-level evaluation is essential because chunk boundaries determine which evidence falls into which unit.

Chunk sizes are defined in tokens of a single reference tokenizer (`bert-base-uncased` WordPiece), so that identical chunk sets are embedded by every dense model. This isolates embedding-model effects (RQ4) from segmentation effects: differences across embedders reflect only the encoder, not divergent segmentations.

3.2. Dataset and Relevance Labels

We use QASPER [6], a corpus of 1,585 scientific NLP papers (extracted from S2ORC [24]) with 5,049 information-seeking questions written by practitioners reading only the title and abstract, and answered—with supporting

[1]Repository with experiment code, pinned dependencies, filtered data manifest, and all raw result files; the analysis plan in the versioned specification was fixed before outcome data were examined, with deviations documented inline. The complete archive is available from the author upon request during review.

Table 1: Experimental factors. Dense conditions use the full factorial of strategy × size × overlap per embedder (27 configurations per embedder); BM25 uses the 24 windowing configurations, since semantic segmentation is embedder-defined (Section 3.4).

Factor	Levels	Notes
Strategy	fixed-token, sentence-window, recursive-split, structure-aware, semantic	5 levels
Chunk size	128 / 256 / 512 tokens	reference tokenizer
Overlap	0% / 20% of size	none for semantic
Embedder	MiniLM-L6, BGE-small	dense only
Retriever	dense cosine / BM25	BM25: no embedder

evidence spans—by separate annotators who read the full text. Evidence spans provide human chunk-level relevance judgments without any synthetic or model-generated labeling, unlike recent LLM-judged evaluations [4].

From the QASPER train split we apply fixed filters, in order: (i) retain papers whose full text parses to more than 2,000 reference tokens; (ii) retain questions having at least one answerable extractive-span reference answer (dropping yes/no and free-form-only questions); (iii) drop questions whose largest reference's evidence exceeds 30% of document length (degenerate supervision); (iv) drop papers left with zero usable questions. The first 60 surviving papers by stable identifier order form the study sample (120 question–paper pairs). Gold spans are located by whitespace-normalized substring search into the assembled document text; spans that cannot be located (0/233 in our sample) would be counted and excluded per question. Figure 1 illustrates the object of study: the same document segmented by two strategies for one real question, with annotated evidence highlighted; boundary placement decides whether evidence arrives concentrated in few chunks or fragmented across many.

3.3. Experimental Design

The design was planned as a full factorial over five strategies, three sizes, two overlaps, and three embedders. Two scope adjustments, made before any analysis, are documented here as deviations from the initial plan. First, the third embedder (E5-small [25]) was dropped when measured throughput on the available CPU projected beyond the experimental budget; the

pre-registered contingency plan specified exactly this drop order. Second, semantic chunking is excluded from sparse retrieval because its boundaries depend on an embedder's sentence representations. The final grid therefore contains $27 \times 2 = 54$ dense configuration–embedder combinations plus 24 sparse ones. RQ4 is scoped accordingly: stability is assessed across the two retained embedders and across dense-versus-sparse ranking. Within the realized budget the BGE-small arm covers the first 30 papers of the sample (stable identifier order); all BGE statistics are paired within that subset, per the pre-registered contingency plan.

3.4. Chunking Strategies

All strategies receive the parsed document (section boundaries, sentence spans, reference-token offsets) and emit ordered, non-overlapping-in-list chunks covering at least 95% of document characters.

Fixed-token windowing. A sliding window of z reference tokens with step $z - r$, where $r = \lfloor 0.2z \rfloor$ when overlap is enabled, else $r = z$. With overlap disabled the tiling is exactly disjoint and the final window may be shorter than z.

Sentence-window. Sentences are detected by a boundary regex (period/exclamation/question mark followed by whitespace and an uppercase, quote or parenthesis character), merging fragments under 20 characters into their predecessor. Windows of k sentences are formed, where $k = \mathrm{round}(z/\bar{\ell})$ and $\bar{\ell}$ is the mean sentence token length of the document; step $k - \mathrm{round}(kr)$ analogously.

Recursive splitting. Following the hierarchical heuristic popularized by LangChain [11], text is split on paragraph breaks, then line breaks, then sentence boundaries, then words; atomic pieces are packed greedily into groups of at most z tokens, and on overflow the next group is seeded with trailing pieces totaling approximately r tokens.

Structure-aware. Sections are treated as atomic super-units, packed whole while they fit within $1.15z$ tokens; an oversize section is packed internally by the recursive packer, so chunks never cross section boundaries and never overlap across them.

Semantic. Every sentence of the document is embedded by the configuration's embedder; a breakpoint is placed between adjacent sentences whose cosine similarity falls below $\tau = 0.5$ [20]; chunks smaller than half the target size are merged forward (the last may merge backward); chunks are never

split. Because breakpoints depend on sentence embeddings, each dense embedder induces its own segmentation, and no segmentation exists for BM25.

3.5. Retrieval Configuration

Dense retrieval embeds chunks and queries with one of two sentence encoders—`all-MiniLM-L6-v2` [26, 8] and `bge-small-en-v1.5` [27]—using normalized embeddings and dot-product scoring. No model-specific instruction prefixes are applied to either queries or chunks: uniform treatment keeps the protocol identical across models, which the stability comparison requires. Both encoders run through `sentence-transformers` with default sequence limits, so over-length inputs are wordpiece-truncated at 256 tokens for MiniLM-L6-v2 and 512 for BGE-small-en-v1.5 rather than split-and-pooled; truncation bites most in coarse configurations (Section 5). Sparse retrieval uses BM25 (Okapi, 28) over lowercased alphanumeric tokens, with the `rank_bm25` reference implementation defaults ($k_1 = 1.5$, $b = 0.75$). Per paper, the index consists of that paper's chunks and the queries are its questions; rankings are thus document-local, matching how per-document RAG indexes operate.

3.6. Metrics

A retrieved chunk is *relevant* iff its span strictly overlaps at least one gold evidence span. For graded evaluation we count distinct gold spans overlapped, capped at three, yielding relevance grades $g \in \{0, 1, 2, 3\}$ used only by nDCG [21]. We report Recall@k ($k \in \{5, 10, 20\}$), MRR, and nDCG@10; Recall@10 serves as the headline recall figure. Metrics are computed per question and averaged within paper; papers are the units of all statistical inference.

3.7. Statistical Analysis Plan

Primary contrasts compare paired per-paper metric deltas within each embedder (and within BM25): (i) size effects 512 vs. 256 and 256 vs. 128 inside each windowing strategy at zero overlap; (ii) overlap effects 20% vs. 0% inside each strategy–size cell; (iii) linguistically informed strategies vs. fixed-token at matched nominal size 256. Because bounded metrics such as Recall@k deviate strongly from normality, deltas are tested with two-sided Wilcoxon signed-rank tests [29]. Families for Holm step-down correction [22] are defined by contrast family × metric, and significance requires adjusted $p < 0.05$. To distinguish meaningful from detectable effects we additionally

require Cliff's delta [23] of at least small magnitude ($|\delta| \geq 0.147$) before calling a contrast significant; contrasts passing the test but not this threshold are reported as "negligible." Headline tables report per-configuration means; medians and dispersion for every configuration are included in the released per-config summary files.

4. Results

We organize findings by research question. Table 2 reports mean nDCG@10 and Recall@10 for every configuration, and Table 3 summarizes contrasts passing the joint criterion; Fig. 2 shows size effects per retriever, Fig. 3 paired overlap effects, and Fig. 4 the strategy–size landscape. Unless stated otherwise, "significant" means this joint criterion; correction families are defined within retriever, contrast family, and metric.

4.1. RQ1: Chunk Size

Chunk size is the strongest and most consistent factor we measured. Every windowing strategy improves monotonically with size across retrievers (Fig. 2). On MiniLM-L6 dense retrieval, moving from 128- to 256-token chunks raises mean nDCG@10 by +0.105 for fixed-token windows ($\delta = 0.25$, $p^{\text{Holm}} = 2 \times 10^{-4}$, $n = 60$ papers); moving from 256 to 512 adds +0.100. The pattern replicates exactly on BM25, where 512 vs. 256 yields +0.150 nDCG@10 ($\delta = 0.36$): larger chunks concentrate more of each question's evidence into single units, benefiting lexical and dense ranking alike. On the headline nDCG@10 metric, fifteen of the sixteen size contrasts across strategies and retrievers are significant with at least small effects; the sole exception (structure-aware, 512 vs. 256, MiniLM) is significant with negligible effect size. No size contrast is negative for any metric in any configuration.

4.2. RQ2: Chunk Overlap

Overlap never helps: across all 108 overlap contrasts on Recall@k (4 strategies × 3 sizes × 3 metrics × 3 retrievers; semantic chunking has no overlap dimension), no contrast is significantly positive anywhere in the grid. Twenty-five contrasts are significantly *negative* after Holm correction, exclusively on Recall@k (12 at k=10, 9 at k=20, 4 at k=5), spanning all three retrievers (Fig. 3); a further 14 are significant with negligible effect sizes ($|\delta| < 0.147$), for 39 in total. nDCG@10 and MRR are overwhelmingly unaffected (23/24 and 24/24 null at the cell level). The asymmetry is partly

mechanical: overlapping windows duplicate evidence-bearing content, inflating the number of relevant chunks that Recall@k normalizes against while duplicated copies occupy rank slots without adding unique evidence. Both effects push Recall down at fixed k. Denominator-insensitive rank-position metrics show no offsetting gain. Under document-local indexing, the common default of 20% overlap therefore costs up to 25% additional index entries and measurably degrades top-k evidence coverage.

4.3. RQ3: Strategy Comparisons

The three linguistically informed strategies do not outperform fixed-token windows at matched nominal size 256: their contrasts are not significant for any metric on MiniLM-L6, and on BM25 the sole significant contrast is *negative* (structure-aware loses -0.101 Recall@5, $\delta = -0.17$). Differences among windowing strategies at 512 tokens are likewise not significant on BM25. The decisive exception is *semantic* chunking under dense retrieval, which is significantly better than fixed-token windows at every matched size, with effects that grow with size: $+0.120$ ($\delta = 0.26$), $+0.192$ ($\delta = 0.43$), and $+0.318$ ($\delta = 0.65$) nDCG@10 at nominal sizes 128, 256, and 512 respectively (all $p^{\text{Holm}} \leq 10^{-3}$). Because semantic chunking was not part of pre-specified contrast family (iii) in Section 3.7, we report these size-generalized contrasts as *exploratory*, Holm-corrected within their own family. In absolute terms, semantic chunks of nominal size 512 reach mean nDCG@10 of 0.863 versus 0.559 for the best windowing strategy, and even nominal-256 semantic chunks (0.635) outperform every 512-token windowed configuration.

Two qualifications sharpen this finding. First, semantic breakpoints are data-driven rather than size-driven: at nominal size 512 a semantic segmentation averages only about 2–6 chunks per document versus about 15 for windowed configurations, so part of its headline advantage reflects coarser units. At matched nominal sizes the advantage remains significant and grows with size ($+0.12$/$+0.19$/$+0.32$ nDCG@10 at 128/256/512), but at equal index granularity it shrinks to statistical equality: semantic-128 versus fixed-token-256 (28.0 vs. 29.1 chunks per document) yields only $+0.015$ nDCG@10 ($p = 0.97$); both granularity-matched pairings are post hoc. The honest summary is that topic-coherent coarse segmentation - not boundary placement alone - carries the gain: semantic-256 significantly beats even fixed-token-512 ($+0.091$, $p = 0.018$, exploratory) while indexing about 40% fewer units. Second, semantics-aware segmentation is also cheaper to serve: relative to fixed-token windows at size 512 it shrinks the dense index by about 80%; its

extra cost is confined to one sentence-level embedding pass per document before chunk embedding. This mechanism - cutting where the embedder itself detects topic shifts - requires an embedder-aware segmenter, which explains why no analogous gain appears under BM25.

4.4. RQ4: Stability

Within the compute budget, the BGE-small arm covers the first 30 of the 60 papers (stable identifier order); all comparisons in this subsection are paired within that subset, while every other subsection uses all 60 papers. Stability is strong. Configuration rankings by mean nDCG@10 correlate across embedders at Spearman $\rho = 0.93$ ($p = 2 \times 10^{-12}$, $n = 27$ configurations); because the BGE ranking is estimated on the 30-paper subset, this coefficient also absorbs subsample noise. Both encoders select the same best configuration (semantic, nominal size 512). The semantic advantage of Section 4.3 fully replicates: under BGE-small, semantic chunks beat fixed-token windows significantly at every matched size (+0.321, +0.288, +0.132 nDCG@10 for sizes 128/256/512; δ up to 0.64; all $p^{\text{Holm}} \leq 0.016$). Of the 130 shared statistical contrasts, verdicts agree on 77.7% (35 significant in both arms); the remainder split toward MiniLM-L6 (23 MiniLM-only, 6 BGE-only), consistent with the reduced BGE sample, and size-contrast detection agrees on 5 of 8 nDCG@10 pairs. Dense-versus- sparse consistency of the two dominant effects—large size gains and no overlap benefit—is already established on the full sample in Sections 4.1 and 4.2.

4.5. Error Analysis

Chunking quality bounds retrieval from above: taking the best configuration per question, 100 of the 120 questions attain a perfect nDCG@10 under some MiniLM-L6 configuration (median best-case = 1.00; 109 of 120 across all retrievers), so most retrieval failures observed in any single configuration are induced by segmentation choices rather than by missing evidence. Conversely, per-question performance varies strongly across configurations, confirming that boundary placement decides which questions are answerable at rank 10. The mechanism is a coverage–dilution trade-off: larger chunks gather an answer's scattered evidence into one chunk but embed more competing content, which can push that chunk below rank 10 for narrow quantitative questions; smaller chunks isolate such details but fragment multi-span evidence. Semantic boundaries mitigate precisely this trade-off

by cutting where topical coherence breaks, consistent with their advantage in Section 4.3.

5. Discussion

5.1. Why Size Helps and Overlap Does Not

The size effect (Section 4.1) follows from evidence granularity: QASPER questions are answered by spans whose supporting context extends over multiple sentences, so larger chunks more often contain a question's complete evidence set in one unit, lifting both recall and early-rank precision. Overlap is not merely unhelpful but measurably harmful to Recall@k (Section 4.2). Two mechanisms combine. First, duplicated boundary content consumes rank slots with near-identical copies, crowding out unique evidence at fixed k. Second, duplication inflates the number of relevant chunks that Recall normalizes against, so the same absolute evidence coverage yields a lower score; we flag this denominator coupling as a caveat when interpreting the magnitude of the overlap penalty. Rank-position metrics insensitive to the denominator show no offsetting benefit, which is why nDCG@10 and MRR are largely null rather than positive. Under document-local indexing, overlap adds measurable cost without compensating quality—contrary to widespread default practice.

5.2. Why Semantic Boundaries Win Under Dense Retrieval

Dense retrievers score whole-chunk embeddings; a chunk that mixes topics yields an embedding matching its constituent aspects only weakly. Semantic segmentation places boundaries exactly where adjacent-sentence similarity collapses, yielding internally coherent chunks whose embeddings concentrate mass on one aspect, which is what cosine ranking rewards. The magnitude of the gain (nDCG@10 0.863 vs. 0.559 at nominal size 512) exceeds every other factor we manipulated. Our finding contrasts with Qu et al. [20], who report that semantic chunking rarely justifies its cost; their evaluation relies on stitched or short-document corpora and proxy tasks, whereas scientific papers carry strong section structure with sharp topic shifts—precisely the regime where similarity breakpoints align with argumentative units. That this advantage appears under human labels but not proxy tasks suggests a testable hypothesis: semantic chunking pays off in proportion to how far a document's internal topic boundaries diverge from uniform spacing. A mirror-image mechanism likely explains why structure-aware chunking never

beats fixed-token windows and is the only strategy significantly worse than them on BM25: QASPER evidence spans frequently straddle section boundaries, so hard section-aligned cuts can split one evidence passage across two chunks—each partially relevant, both competing for rank slots—whereas a size-driven cut has no stake in where sections end. Structure-aware segmentation should pay off only when evidence units align with the markup, a corpus property worth measuring before deploying it.

5.3. Practical Guidance

For implementers of per-document RAG indexes over long structured text, our results support three rules of thumb. First, exhaust chunk size before strategy tuning: moving 128→512 tokens yielded gains no other lever matched. Second, drop overlap defaults: they inflate the index by up to 25% and never helped any metric. Third, where documents are topically heterogeneous, invest one sentence-level embedding pass in similarity-based segmentation: it outperformed every linguistically informed strategy and shrank the dense index by about 80% relative to fixed-token windows; boundary quality, not extra index size, drives the gain.

5.4. Threats to Validity

Construct. Binary relevance from span overlap treats a chunk as relevant even when it contains one evidence sentence among much unrelated content; graded nDCG partially mitigates this, and both views agree in direction across all contrasts. Locating gold spans by normalized substring search can miss paraphrase-offset annotations; none failed in our sample (0/233 spans). **Internal.** Semantic chunking changes effective chunk counts as well as boundaries, confounding strategy with index granularity; we mitigate by matching granularity explicitly (Section 4.3) and note the residual confound that semantic chunks at nominal size 512 are far coarser than windowed ones. The semantic similarity threshold $\tau = 0.5$ is inherited from prior work [20] without a sensitivity sweep; other thresholds may shift the crossover between semantic and windowed strategies. Reference-tokenizer choice fixes segmentation across embedders but means absolute token counts differ from each embedder's native tokenizer. Because over-length chunks are truncated to each encoder's sequence limit rather than pooled (Section 3.5), coarse-window dense results partly reflect truncated representations; the direction of the size and semantic effects is unlikely to reverse under pooling, but absolute levels should be read with this in mind. Recall@k normalizes by the

number of relevant chunks in the index, which overlap duplication inflates; the negative overlap effects on Recall therefore partly reflect this denominator coupling, and we rely on the null (never positive) rank-position metrics to support the practical conclusion. Finally, chunkings cover at least 95% of document characters by design; the uncovered remainder is duplicated boundary whitespace and heading seams, which contain no evidence spans in our sample. **External.** One domain (NLP scientific papers), one language, two small embedders, document-local retrieval without reranking; results may differ for short documents, cross-document corpora, instruction-tuned embedders, or hybrid rankers. RQ4 stability claims are correspondingly scoped. **Statistical.** A Monte Carlo analysis of the paired design (4,000 replicates, $n = 60$) puts the minimum detectable mean delta at 80% power at approximately 0.121 nDCG@10, 0.185 Recall@10, and 0.149 MRR — commensurate with the pre-registered small-effect bar ($|\delta| \geq 0.147$). True effects below those magnitudes should be read as "not detected," not "absent." The paper-level pairing assumes questions within a paper are exchangeable; metric averaging within paper limits their leverage accordingly.

6. Conclusion

We presented a controlled evaluation with a pre-specified analysis of five chunking strategies for dense and sparse retrieval over long scientific documents with human evidence labels. Three findings are robust across metrics, strategies, embedders, and sparse ranking: retrieval quality improves substantially as chunks grow to 512 reference tokens, the largest size tested; chunk overlap adds cost without measurable benefit and often reduces top-k coverage; and among strategies, embedding-based semantic segmentation dominates—by a wide margin on dense retrieval—while the linguistically informed, semantics-free strategies are statistically indistinguishable from fixed-token windows. Because nearly every question was perfectly recoverable under at least one configuration (109 of 120 across retrievers), these differences reflect segmentation quality directly. Future work should extend the factorial design to additional domains and embedders, sweep the semantic threshold τ, evaluate adaptive-size segmenters against the coverage–dilution trade-off identified here, and couple chunk-level retrieval quality to end-to-end answer generation to measure how retrieval gains propagate downstream.

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Md. Mahin Ahmed: Conceptualization, Methodology, Software, Investigation, Data curation, Writing – original draft, Writing – review & editing.

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Data availability

The analysis code, frozen configuration, corpus filters, cached model outputs, and every result file underlying the tables and figures are available from the author upon request and will be deposited in a public repository with a permanent DOI upon acceptance.

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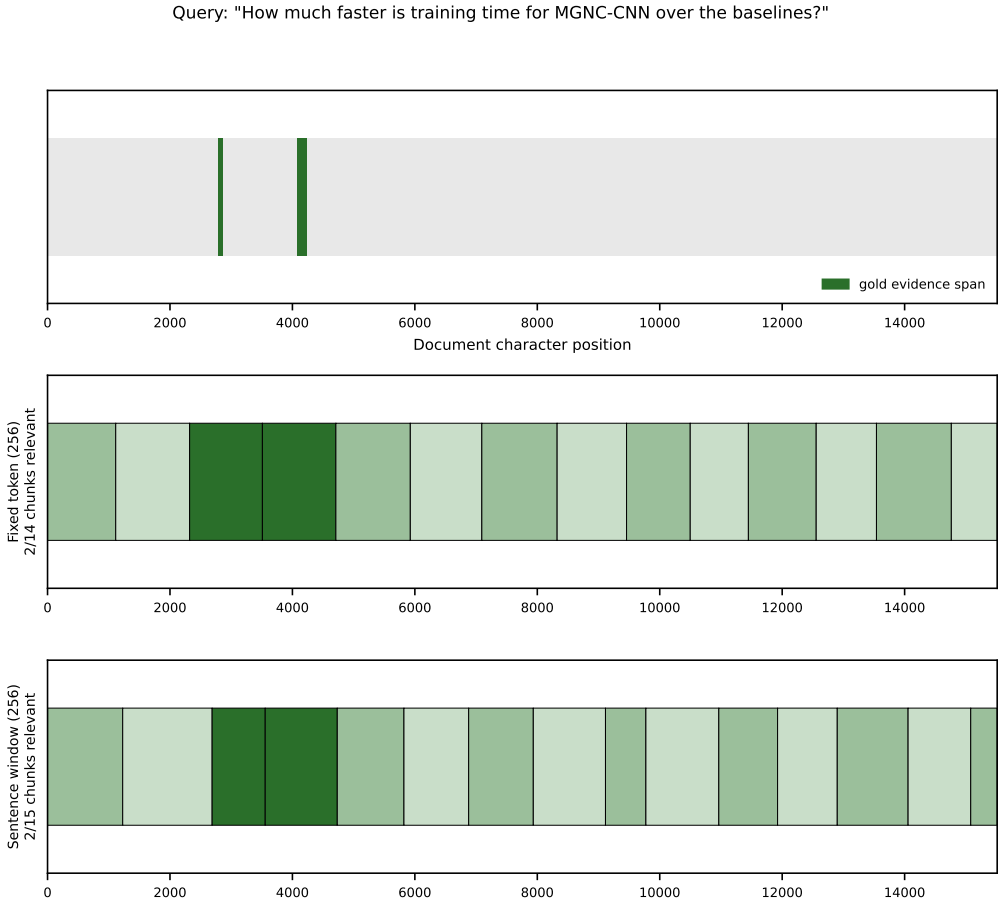


Figure 1: One QASPER question against the same paper under two chunking strategies. Top: document extent with human-annotated evidence spans in green. Rows below: chunks produced by fixed-token and sentence-window segmentation at nominal size 256; dark green chunks overlap gold evidence.

Table 2: Mean nDCG@10 and Recall@10 for every chunking configuration, grouped by retriever. BGE-small covers the first 30 of 60 papers; BM25 has no semantic rows (see Section 3.3).

Retriever	Strategy	Size	Overlap	nDCG@10	Recall@10
MiniLM-L6	Fixed	128	0%	0.339	0.596
MiniLM-L6	Fixed	128	20%	0.325	0.543
MiniLM-L6	Fixed	256	0%	0.444	0.730
MiniLM-L6	Fixed	256	20%	0.428	0.642
MiniLM-L6	Fixed	512	0%	0.544	0.894
MiniLM-L6	Fixed	512	20%	0.516	0.831
MiniLM-L6	Sentence	128	0%	0.326	0.592
MiniLM-L6	Sentence	128	20%	0.309	0.519
MiniLM-L6	Sentence	256	0%	0.444	0.746
MiniLM-L6	Sentence	256	20%	0.393	0.636
MiniLM-L6	Sentence	512	0%	0.555	0.945
MiniLM-L6	Sentence	512	20%	0.544	0.907
MiniLM-L6	Recursive	128	0%	0.337	0.568
MiniLM-L6	Recursive	128	20%	0.315	0.408
MiniLM-L6	Recursive	256	0%	0.418	0.691
MiniLM-L6	Recursive	256	20%	0.376	0.575
MiniLM-L6	Recursive	512	0%	0.559	0.910
MiniLM-L6	Recursive	512	20%	0.533	0.839
MiniLM-L6	Structure	128	0%	0.343	0.540
MiniLM-L6	Structure	128	20%	0.296	0.443
MiniLM-L6	Structure	256	0%	0.431	0.719
MiniLM-L6	Structure	256	20%	0.397	0.605
MiniLM-L6	Structure	512	0%	0.493	0.818
MiniLM-L6	Structure	512	20%	0.490	0.825
MiniLM-L6	Semantic	128	0%	0.459	0.758
MiniLM-L6	Semantic	256	0%	0.635	0.944
MiniLM-L6	Semantic	512	0%	0.863	1.000
BGE-small	Fixed	128	0%	0.356	0.591
BGE-small	Fixed	128	20%	0.283	0.467
BGE-small	Fixed	256	0%	0.410	0.741
BGE-small	Fixed	256	20%	0.365	0.562
BGE-small	Fixed	512	0%	0.585	0.911
BGE-small	Fixed	512	20%	0.567	0.843
BGE-small	Sentence	128	0%	0.291	0.500
BGE-small	Sentence	128	20%	0.263	0.380
BGE-small	Sentence	256	0%	0.358	0.673
BGE-small	Sentence	256	20%	0.335	0.558
BGE-small	Sentence	512	0%	0.556	0.894
BGE-small	Sentence	512	20%	0.556	0.831
BGE-small	Recursive	128	0%	0.314	0.601
BGE-small	Recursive	128	20%	0.312	0.409
BGE-small	Recursive	256	0%	0.366	0.663

Table 3: Contrast families passing the joint criterion (Holm-adjusted $p < 0.05$ with $|\delta| \geq 0.147$) in at least one retriever. Ranges span retrievers within each family × contrast × metric cell; n is the minimum per-retriever sample size. † exploratory contrast, not part of the pre-specified analysis plan (Section 3.7); Holm-corrected within its own family.

Metric	Family	Contrast	Δ range	δ range	n	$p_{\max}^{\text{Holm}}$
mrr	fixedtok size	t256-t128	[+0.073, +0.079]	[0.19, 0.20]	60	0.0004
mrr	fixedtok size	t512-t256	[+0.126, +0.173]	[0.26, 0.32]	30	0.0005
mrr	recursive size	t256-t128	[+0.059, +0.066]	[0.17, 0.18]	60	0.0192
mrr	recursive size	t512-t256	[+0.099, +0.169]	[0.25, 0.33]	30	0.0039
mrr	semantic vs-fixedtok†	s128-vs-f128	[+0.085, +0.278]	[0.21, 0.53]	30	0.0103
mrr	semantic vs-fixedtok†	s256-vs-f256	[+0.163, +0.298]	[0.33, 0.56]	30	0.002
mrr	semantic vs-fixedtok†	s512-vs-f512	[+0.137, +0.355]	[0.27, 0.61]	30	0.045
mrr	sentwin size	t256-t128	[+0.046, +0.091]	[0.17, 0.22]	60	0.0057
mrr	sentwin size	t512-t256	[+0.074, +0.182]	[0.20, 0.34]	30	0.0121
mrr	struct size	t256-t128	[+0.055, +0.055]	[0.19, 0.19]	60	0.0005
mrr	struct size	t512-t256	[+0.095, +0.099]	[0.18, 0.21]	30	0.0399
ndcg@10	fixedtok size	t256-t128	[+0.105, +0.110]	[0.25, 0.26]	60	0.0002
ndcg@10	fixedtok size	t512-t256	[+0.100, +0.174]	[0.20, 0.37]	30	0.0027
ndcg@10	recursive size	t256-t128	[+0.081, +0.116]	[0.20, 0.25]	60	0.0046
ndcg@10	recursive size	t512-t256	[+0.141, +0.205]	[0.31, 0.40]	30	0.0005
ndcg@10	semantic vs-fixedtok†	s128-vs-f128	[+0.120, +0.321]	[0.26, 0.64]	30	0.0011
ndcg@10	semantic vs-fixedtok†	s256-vs-f256	[+0.192, +0.288]	[0.43, 0.59]	30	0.0003
ndcg@10	semantic vs-fixedtok†	s512-vs-f512	[+0.132, +0.318]	[0.28, 0.65]	30	0.016
ndcg@10	sentwin size	t256-t128	[+0.077, +0.118]	[0.21, 0.26]	60	0.0008
ndcg@10	sentwin size	t512-t256	[+0.112, +0.198]	[0.24, 0.40]	30	0.0005
ndcg@10	struct size	t256-t128	[+0.089, +0.096]	[0.19, 0.21]	60	0.0019
ndcg@10	struct size	t512-t256	[+0.131, +0.146]	[0.24, 0.28]	30	0.0005
r@10	fixedtok overlap	o20-o00_t256	[-0.092, -0.088]	[-0.17, -0.17]	60	0.0422
r@10	fixedtok size	t256-t128	[+0.134, +0.176]	[0.20, 0.27]	60	0.0028
r@10	fixedtok size	t512-t256	[+0.164, +0.201]	[0.29, 0.36]	30	0.0114
r@10	recursive overlap	o20-o00_t128	[-0.192, -0.161]	[-0.33, -0.23]	30	0.0064
r@10	recursive overlap	o20-o00_t256	[-0.116, -0.116]	[-0.17, -0.17]	60	0.0068
r@10	recursive overlap	o20-o00_t512	[-0.117, -0.071]	[-0.21, -0.17]	30	0.0212
r@10	recursive size	t256-t128	[+0.122, +0.218]	[0.19, 0.34]	60	0.0048
r@10	recursive size	t512-t256	[+0.182, +0.250]	[0.34, 0.42]	30	0.0022
r@10	semantic vs-fixedtok†	s128-vs-f128	[+0.162, +0.368]	[0.23, 0.60]	30	0.0077
r@10	semantic vs-fixedtok†	s256-vs-f256	[+0.214, +0.237]	[0.36, 0.38]	30	0.0091
r@10	semantic vs-fixedtok†	s512-vs-f512	[+0.106, +0.106]	[0.17, 0.17]	60	0.0077
r@10	sentwin overlap	o20-o00_t256	[-0.110, -0.110]	[-0.20, -0.20]	60	0.0057
r@10	sentwin overlap	o20-o00_t512	[-0.064, -0.064]	[-0.15, -0.15]	60	0.0047
r@10	sentwin size	t256-t128	[+0.154, +0.173]	[0.24, 0.26]	30	0.0179
r@10	sentwin size	t512-t256	[+0.194, +0.222]	[0.33, 0.36]	30	0.011
r@10	struct overlap	o20-o00_t128	[-0.170, -0.170]	[-0.28, -0.28]	30	0.0093
r@10	struct overlap	o20-o00_t256	[-0.114, -0.114]	[-0.16, -0.16]	60	0.002
r@10	struct size	t256-t128	[+0.170, +0.178]	[0.25, 0.27]	60	0.0006
r@10	struct size	t512-t256	[+0.099, +0.213]	[0.17, 0.28]	30	0.009
r@10	struct vs-fixedtok	s256-vs-f256	[-0.162, -0.162]	[-0.21, -0.21]	30	0.0092
r@20	fixedtok overlap	o20-o00_t256	[-0.126, -0.062]	[-0.23, -0.21]	30	0.0348
r@20	fixedtok size	t256-t128	[+0.132, +0.244]	[0.29, 0.54]	30	0.0004
r@20	recursive overlap	o20-o00_t128	[-0.157, -0.157]	[-0.28, -0.28]	60	0.001
r@20	recursive size	t256-t128	[+0.121, +0.186]	[0.26, 0.40]	30	0.0281
r@20	recursive size	t512-t256	[+0.097, +0.104]	[0.18, 0.18]	60	0.0032
r@20	semantic vs-fixedtok†	s128-vs-f128	[+0.158, +0.267]	[0.31, 0.60]	30	0.0006
r@20	sentwin overlap	o20-o00_t128	[-0.147, -0.089]	[-0.26, -0.16]	30	0.0142
r@20	sentwin overlap	o20-o00_t256	[-0.068, -0.068]	[-0.18, -0.18]	60	0.0142
r@20	sentwin size	t256-t128	[+0.119, +0.226]	[0.26, 0.42]	30	0.0187
r@20	struct overlap	o20-o00_t128	[-0.169, -0.107]	[-0.26, -0.18]	60	0.0231
r@20	struct size	t256-t128	[+0.124, +0.162]	[0.22, 0.30]	30	0.0351
r@20	struct size	t512-t256	[+0.107, +0.143]	[0.21, 0.24]	30	0.0351
r@20	struct vs-fixedtok	s256-vs-f256	[-0.176, -0.176]	[-0.25, -0.25]	30	0.0112
r@5	fixedtok overlap	o20-o00_t256	[-0.117, -0.117]	[-0.17, -0.17]	60	0.0255
r@5	fixedtok size	t256-t128	[+0.154, +0.158]	[0.23, 0.25]	60	0.0008
r@5	fixedtok size	t512-t256	[+0.140, +0.227]	[0.21, 0.32]	30	0.0054
r@5	recursive overlap	o20-o00_t256	[-0.144, -0.144]	[-0.20, -0.20]	60	0.0044
r@5	recursive overlap	o20-o00_t512	[-0.159, -0.159]	[-0.24, -0.24]	60	0.004
r@5	recursive size	t256-t128	[+0.114, +0.154]	[0.17, 0.22]	60	0.0149
r@5	recursive size	t512-t256	[+0.153, +0.253]	[0.22, 0.37]	30	0.012

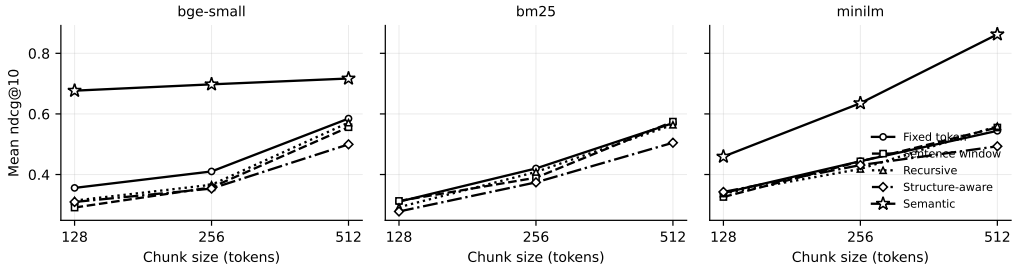


Figure 2: Mean nDCG@10 versus chunk size (overlap 0%; semantic has no overlap condition) per retriever. Larger chunks help every strategy and retriever.

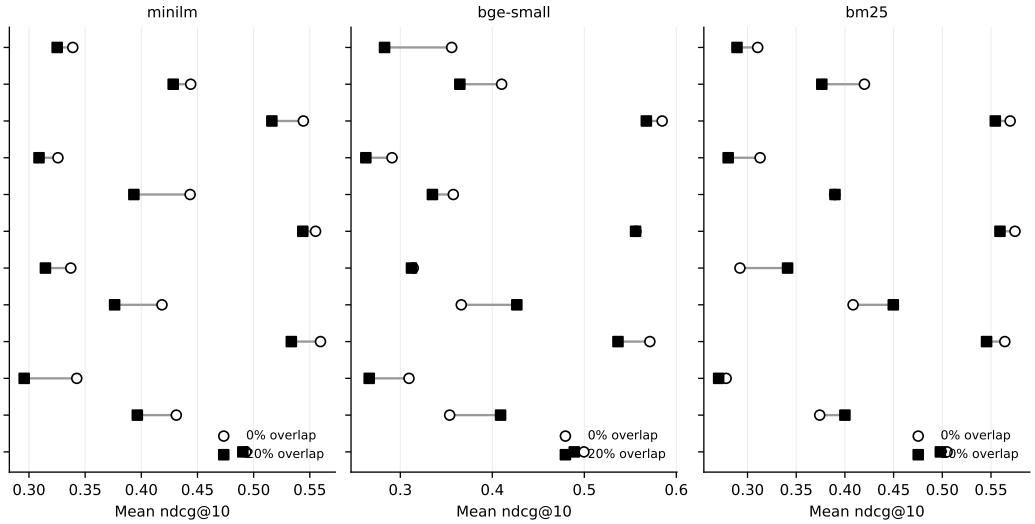


Figure 3: Paired comparison of mean nDCG@10 with 0% versus 20% overlap, per retriever. Circles right of squares indicate overlap helps for that configuration.

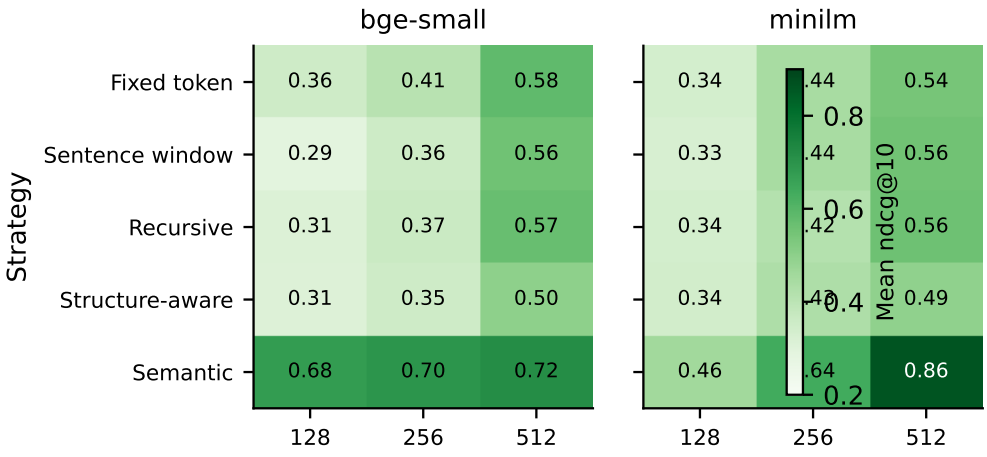


Figure 4: Mean nDCG@10 for strategy × size (overlap 0%) per dense embedder.